ML Project Report

Project Title: Human Activity Recognition using Sensor Data

Students:

1. Sarah Tariq (Roll Number: BSEF23M532)

2. Tehreem Fatima (Roll Number: BSEF23M535)

1. Introduction

In this deliverable, a Fully Connected Neural Network (FCNN) was implemented to classify human activities using the UCI HAR dataset. Instead of raw sensor signals, the engineered features from the previous deliverable were used so the workflow remains consistent.

2. Model Design

A simple neural network was selected to keep the approach understandable. The model consists of three hidden layers with ReLU activation and a softmax output layer for six activity classes. This structure satisfies the assignment requirements while remaining easy to interpret.

3. Training Observation

During training, the loss consistently decreased for both training and validation sets. This indicates that the model was learning meaningful patterns from the features. The validation loss stayed close to training loss, showing no serious overfitting.

4. Results

The FCNN achieved good classification accuracy on the validation set. Activities involving motion such as walking were predicted more accurately, while some confusion was observed between static activities like sitting and standing. This behavior is expected due to their similar motion characteristics.

The F1-scores show that dynamic activities like walking or walking upstairs were predicted really well, while static activities like sitting and standing were a bit tricky because of their similarity.

5. Comparison with Classical ML

Compared to the Random Forest model from Deliverable 5.2, the neural network performed competitively but did not significantly outperform it. This shows that handcrafted features work well with classical models, while neural networks benefit more from raw sequential data.

6. Conclusion

This deliverable successfully extends the project by introducing a neural network model. The FCNN provides a clean and understandable baseline for deep learning approaches and sets the stage for future work using CNNs or LSTMs on raw sensor signals. Even though this model works well with the features we have, it might not do as great on raw sensor data - using CNNs or LSTMs could help with that in the future.